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PAPER

nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

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Abstract

Motivation: Reconstructing eukaryotic viral genomes from metagenomic data is challenging due to their extensive diversity and potential genome segmentation. Current approaches often rely on labor-intensive manual curation for reference selection and scaffolding, limiting scalability for large studies or rapid outbreak response. We address the critical need for an automated, scalable pipeline for efficient viral metagenomic analysis without manual intervention.

Results: We present nf-core/viralmetagenome, a comprehensive Nextflow pipeline for the untargeted reconstruction and variant analysis of eukaryotic DNA and RNA viruses from short-read metagenomic or hybridisation capture enriched samples. The pipeline automates the entire process from read preprocessing to consensus generation, integrating multiple *de novo* assemblers, automated reference selection, and iterative consensus refinement. It features robust quality control, extensive documentation, and seamless portability via Docker and Singularity. We validated the pipeline on diverse simulated and real datasets, demonstrating its ability to recover high-quality genomes from complex metagenomic samples and resolve co-infections, making it a powerful tool for viral surveillance.

Availability: nf-core/viralmetagenome is freely available at https://github.com/nf-core/viralmetagenome with comprehensive documentation at https://nf-co.re/viralmetagenome

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Supplementary information: Supplementary data are available at https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript online.

Key words: viralmetagenome, bioinformatic pipeline, nextflow, viral metagenomics, viral assembly, viral variant analysis

Introduction

Reconstructing viral genomes from metagenomic sequencing data presents considerable computational challenges, particularly for viruses exhibiting extensive genetic diversity (Baaijens et al. 2017, Meleshko et al. 2021). This challenge is amplified in segmented viruses such as influenza, rotavirus, and bunyaviruses, where individual segments can evolve under distinct selective pressures and reassortment complicates reconstruction. While some pipelines target specific viruses and their subtypes (Shepard et al. 2016), accurate genome reconstruction from samples with unknown viral strains typically requires time-consuming manual curation of contigs and reference matching (de Vries et al. 2021). This makes large-scale metagenome studies or rapid outbreak responses impractical.

To address these limitations, we developed nf-core/viralmetagenome, a comprehensive pipeline specifically designed for untargeted viral genome reconstruction. The pipeline is developed using Nextflow (Di Tommaso et al. 2017) within the nf-core framework (Ewels et al. 2020, Langer et al. 2025), ensuring reproducibility through containerization with Docker (Merkel 2014) and Singularity (Kurtzer et al. 2017), and enabling portability across computational platforms such as local desktops, high-performance clusters and cloud environments.

Pipeline Description

nf-core/viralmetagenome implements an automated workflow for *de novo* assembly, reference matching, and iterative consensus refinement to reconstruct viral genomes without prior target knowledge. The pipeline comprises five major stages: read preprocessing, metagenomic diversity assessment, contig assembly and scaffolding, iterative consensus refinement with variant analysis, and quality control (Figure 1). Unless

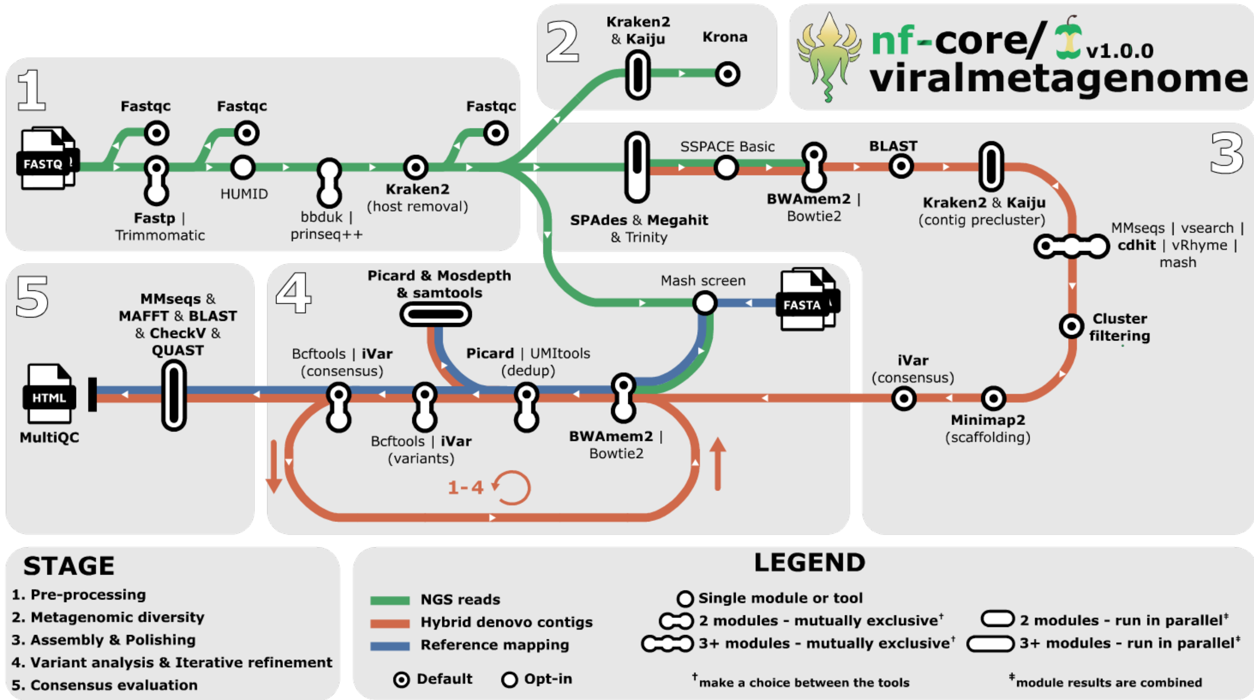

Fig. 1. Overview of the nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction. Default tools are highlighted in bold. nf-core/viralmetagenome processes short-read data through pre-processing, metagenomic diversity assessment, de novo assembly with multiple assemblers, scaffolding with automated reference identification and taxonomy-guided clustering of contigs, and iterative consensus refinement through read mapping and variant calling. Quality control metrics, assembly statistics, and coverage data are integrated into interactive MultiQC reports and standardised overview tables for downstream analysis.

otherwise specified, the pipeline offers multiple options to accommodate established user workflows and preferences. Detailed tool descriptions are provided in Supplementary Table 1.

Read preprocessing

Input reads are provided via sample sheets containing sample names and paths to short-read FASTQ files. These are preprocessed using FastQC and trimmed for adapters using fastp (Chen et al. 2018) (default) or Trimmomatic (Bolger et al. 2014). Fastp offers automated adapter detection and faster processing (Chen et al. 2018). UMI deduplication is implemented using HUMID (Laros) for raw reads, or UMI-tools after mapping (Smith et al. 2017). Multiple sequencing runs can be merged after trimming by specifying merge group identifiers in the sample sheet. Filtering with bbduk (Bushnell 2014) or PRINSEQ++ (Cantu et al. 2019) removes low-complexity sequences, while host reads are removed with Kraken2 (Wood et al. 2019).

Metagenomic diversity assessment

Taxonomic classification of preprocessed reads is performed using two complementary approaches - Kaiju (Menzel et al. 2016) and Kraken2 (Wood et al. 2019) - to maximise detection sensitivity across diverse viral families. Results from both classifiers are visualised using Krona (Ondov et al. 2011).

De novo assembly and clustering

The assembly workflow employs a multi-assembler approach where contigs generated by all tools are pooled and processed jointly. *De novo* assembly uses SPAdes (Meleshko et al. 2021) (RNAviral mode), MEGAHIT (Li et al. 2016), and Trinity (Grabherr et al. 2011), capitalizing on distinct algorithms to maximise genome recovery across diverse viral families and read depths. Subsequently, a BLASTn (Altschul et al. 1990) search against the Reference Viral Database (RVDB) (Goodacre et al. 2018) identifies candidate references for each contig. All contigs and identified references are then subjected to clustering to group related sequences regardless of their assembly origin. This strategy enables contig scaffolding against the cluster centroid—whether an external reference or a *de novo* contig—thereby combining outputs from multiple assemblers into unified scaffolds.

Clustering is performed in two sequential stages. First, taxonomic pre-clustering groups contigs based on taxonomic classification using Kraken2 (Wood et al. 2019) and Kaiju (Menzel et al. 2016), with optional filtering to focus on specific taxonomic clades for more targeted analyses. Second, nucleotide similarity clustering within taxonomic groups is performed using CD-HIT-EST (Li and Godzik 2006) (default), VSEARCH (Rognes et al. 2016), MMseqs2 (Steinegger and Söding 2017), vRhyme (Kieft et al. 2022), or Mash (Ondov et al. 2019). All tools are valid options, though performance may vary depending on the dataset (Zielezinski et al. 2025, Steinegger and Söding 2017).

Optionally, following assembly and extension, reads can be mapped to all contigs using BWA-MEM2 (Vasimuddin et al.

2019) (default), or Bowtie2 (Langmead et al. 2019). Clusters are filtered based on the total percentage of reads mapping to all contigs within a cluster, enabling the removal of likely assembly artefacts. For the final scaffolding step, all cluster members are mapped to the cluster representative or centroid using Minimap2 (Li 2018), followed by consensus calling with iVar (Grubaugh et al. 2019) to generate reference-assisted assemblies.

Iterative consensus refinement and variant calling

The consensus module supports external reference-based analysis and scaffold refinement. Users can provide specific reference genomes; if a set is provided, Mash (Ondov et al. 2019) selects the most similar one. The selected reference genome undergoes a single round of refinement to generate the consensus.

In contrast, scaffold refinement employs an iterative approach, performing up to 4 iterative cycles (default 2) to polish the consensus sequence. Both processes map reads using BWA-MEM2 (Vasimuddin et al. 2019), or Bowtie2 (Langmead et al. 2019), followed by variant calling with BCFtools (Danecek et al. 2021) or iVar (Grubaugh et al. 2019). Benchmarking (Bassano et al. 2022) indicated that BCFtools outperforms iVar in precision and recall, while iVar identified more low-frequency variants. Users can prioritise sensitivity or specificity when selecting the variant caller. Final variant calling is performed on the polished consensus to characterise single nucleotide variants.

Consensus Quality Control

Quality control employs CheckV (Nayfach et al. 2021) to assess completeness, BLASTn (Altschul et al. 1990) for reference similarity, and MMseqs2 (Steinegger and Söding 2017) against the annotated database Virosaurus (Gleizes et al. 2020). These analyses enable species identification, genomic segment classification, host prediction, and retrieval of additional metadata embedded within the reference databases.

The refinement progression is evaluated through sequence alignment with MAFFT (Katoh et al. 2002), which compares final consensus genomes against *de novo* contigs, intermediate consensus sequences from iterative cycles, and the scaffolding reference. All tool metrics are compiled into an interactive MultiQC report (Ewels et al. 2016). Additionally, key metrics are extracted from the MultiQC report and compiled into standalone overview tables to facilitate downstream analysis across all processed samples.

Applications

We validated the pipeline's performance using a diverse set of real metagenomic samples containing human and plant pathogens. nf-core/viralmetagenome successfully reconstructed high-quality or near-complete consensus genomes across various viral families, including segmented viruses (Lassa virus, Orthonairovirus, Tomato spotted wilt tospovirus) and non-segmented viruses (SARS-CoV-2, West Nile virus, Potato virus Y, Youcai mosaic virus, and Monkeypox virus; Supplementary methods S1). The pipeline proved efficient, processing 28 samples in 412 CPU hours with a maximum memory footprint of 79GB on a high-performance cluster. To facilitate local deployment, we implemented a `-profile local` option, which utilizes a smaller viral RefSeq database instead of the full RVDB database for Kaiju, reducing the pipeline's peak RAM usage to approximately 18GB.

To evaluate the pipeline's robustness in complex scenarios, we conducted two targeted experiments. First, we assessed its ability to resolve co-infections using a simulated mixture of viral genomes (Supplementary Methods S2). The pipeline successfully distinguished and reconstructed distinct coinfecting strains, provided the genetic divergence was sufficient ($\leq$ 88.7% ANI; Supplementary Figure 1).

Second, we investigated the critical role of automated reference selection in scaffolding, particularly for segmented viruses with heterogeneous coverage. Using real Lassa virus sequencing data (Supplementary Methods S3), we compared consensus genomes generated using single scaffolding references of varying similarity (64%–100% ANI; supplementary Figure 2) against a baseline consensus reconstruction derived from the full unclustered-RVDB as reference pool. Our analysis revealed that sequencing depth significantly dictates the reliance on reference quality (Figure 2). For the high-coverage S segment (Supplementary Figure S3), *de novo* assembly was sufficient to reconstruct the full genome independently, rendering the choice of scaffolding reference largely redundant (Figure 2A-B). In contrast, for the lower-coverage L segment (Supplementary Figure S3), where *de novo* assembly produced fragmented contigs, the pipeline's ability to identify and utilize a closely related external reference was decisive for bridging gaps and recovering a complete genome (Figure 2C-D).

These findings emphasise the need to keep databases like RVDB (Goodacre et al. 2018) up-to-date. Since nf-core/viralmetagenome is primarily designed for eukaryotic viruses, users interested in bacteriophage analysis are encouraged to explore pipelines designed to target phages such as VIRify (Rangel-Pineros et al. 2022), VIBRANT (Kieft et al. 2020), VirSorter2 (Guo et al. 2021).

Conclusion

nf-core/viralmetagenome addresses a critical need in viral genomics by providing an automated, scalable solution for untargeted viral genome reconstruction. The pipeline successfully automates the traditionally time-consuming and manual process of viral genome reconstruction from short-read metagenomic data through its integrated workflow of *de novo* contig assembly, automated reference selection, clustering algorithms, and iterative refinement strategies.

Our validation results indicate that the pipeline is suitable for a wide range of eukaryotic viral families, it removes the burden of extensive manual curation, and enables consistent, high-quality genome reconstruction with reproducible results across varied computational settings. As viral surveillance and outbreak response increasingly rely on metagenomic sequencing, automated pipelines like nf-core/viralmetagenome will be essential for the timely identification of pathogen strains. The pipeline represents a significant step forward in making viral genome reconstruction accessible to researchers without requiring extensive bioinformatics expertise, facilitating broader adoption of metagenomic approaches in viral research and public health applications.

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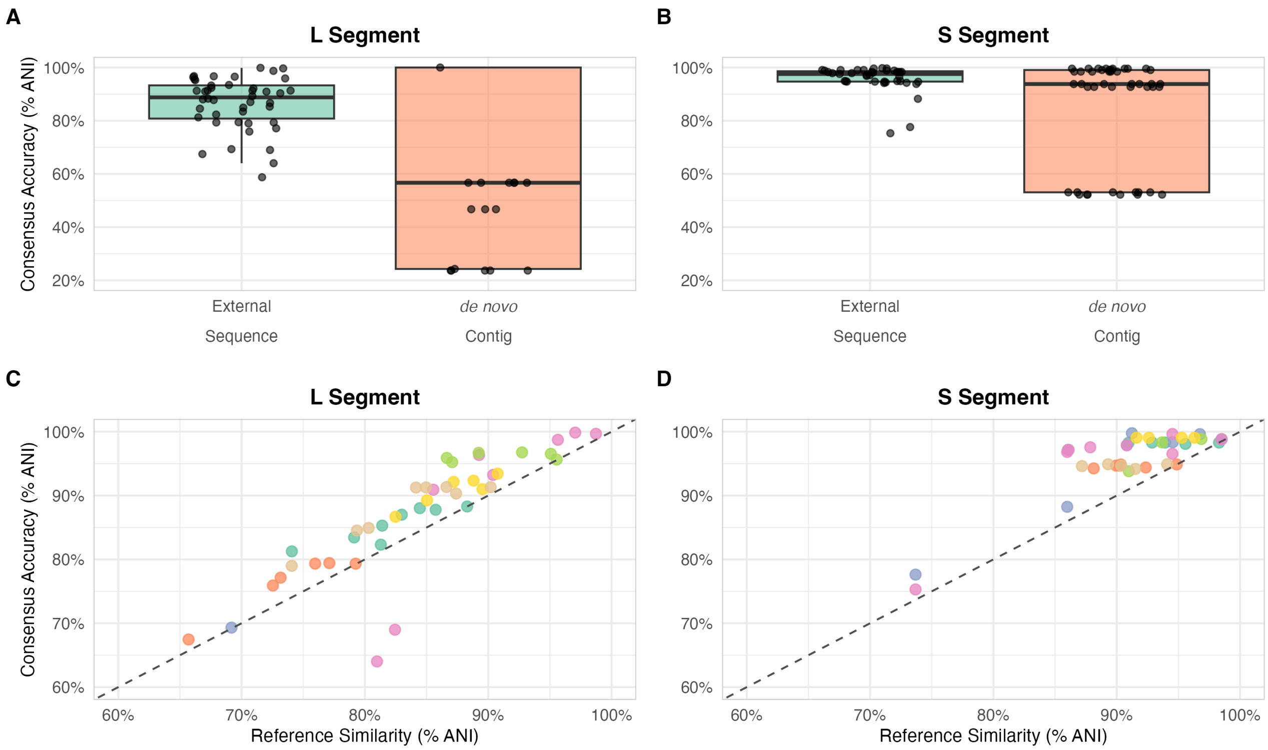

Fig. 2. Impact of scaffolding reference similarity on consensus reconstruction accuracy (n=7 Lassa virus samples). The baseline consensus corresponds to the genome generated using the pipeline's automated reference selection from the complete U-RVDB reference pool. (A, B) Boxplots comparing consensus genome accuracy (ANI % relative to baseline) using external scaffolding reference sequences as versus *de novo* contigs alone for both L and S segments. For the high-coverage S segment, *de novo* contigs perform comparably to external references, whereas for the low-coverage L segment, external references significantly improve accuracy. (C, D) Scatter plots illustrating the relationship between scaffolding reference similarity (ANI % to baseline) and final consensus accuracy (ANI % to baseline) for L and S segments. The dashed line represents the line of identity ($y = x$), and points are colored by sample.

– Vlaanderen, G005323N and G051322N, 1SH2V24N, 12X9222N). Assistance from AI language models (e.g., ChatGPT) was used to enhance the clarity of the manuscript and to support code development.

Author Contributions

J.K. designed and implemented the pipeline, performed validation analyses, and wrote the manuscript. P.L. and L. E. K. supervised the project and provided critical feedback. The nf-core community contributed to maintaining the pipeline. All authors reviewed and approved the final manuscript.

Conflict of Interest

The authors declare no competing interests.

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